

To build the **Causal Engine** of your Diabetes Decision Support System, you need tabular datasets.

For causal inference (using libraries like DoWhy or EconML), you cannot just use any dataset. You specifically need datasets that contain clear **Treatments** (interventions like medication or lifestyle changes), **Outcomes** (like HbA1c levels or readmission), and **Confounders** (like age, BMI, and comorbidities).

Here are the best, production-ready diabetes datasets for your final year project, ranked by how well they fit causal inference tasks.

1. Diabetes 130-US Hospitals Dataset (The Best for Clinical Causal Inference)

This is the most widely used dataset for building clinical models around diabetes. It represents 10 years (1999–2008) of clinical care at 130 US hospitals and integrated delivery networks.

- **Why it's perfect for Causal AI:** It meticulously tracks medication changes. You can ask counterfactual questions like: *"If this patient's insulin had been increased rather than kept steady, would they have avoided hospital readmission?"*
- **The Causal Setup:**
 - **Treatment (T):** Diabetes medications (e.g., Metformin, Insulin) and whether the dosage was up, down, steady, or no.
 - **Outcome (Y):** Hospital readmission within 30 days.
 - **Confounders (W):** Age, race, admission type, number of lab procedures, and primary/secondary diagnoses.
- **Link:** [UCI Machine Learning Repository - Diabetes 130-US Hospitals](#) (Open Access)

2. NHANES (National Health and Nutrition Examination Survey)

Run by the CDC, NHANES combines interviews and physical examinations. It is the absolute gold standard for measuring the causal impact of lifestyle and dietary interventions on physiological markers.

- **Why it's perfect for Causal AI:** It contains highly granular lab data (like actual HbA1c percentages and fasting glucose) combined with exact nutritional intake and prescription drug use.
- **The Causal Setup:**
 - **Treatment (T):** Specific dietary interventions (e.g., reducing carbohydrate intake) or starting a new class of oral hypoglycemics.
 - **Outcome (Y):** HbA1c reduction or fasting blood glucose levels.
 - **Confounders (W):** BMI, socioeconomic status, family history, and physical activity levels.
- **Link:** [CDC NHANES Data Questionnaires and Datasets](#) (Open Access - You can pull specific years, e.g., 2017-2018, which have great diabetes questionnaire modules).

3. CDC Diabetes Health Indicators (BRFSS 2015)

This is a cleaned, machine-learning-ready subset of the Behavioral Risk Factor Surveillance System. It contains over 250,000 survey responses.

- **Why it's perfect for Causal AI:** It is massive, making it great for training complex Double

Machine Learning (DML) or meta-learner models without overfitting. It focuses heavily on macro-level risk factors.

- **The Causal Setup:**
 - **Treatment (T):** Lifestyle changes (e.g., moving from "no physical activity" to "regular physical activity" or reducing BMI category).
 - **Outcome (Y):** Progression from pre-diabetes to Type 2 Diabetes.
 - **Confounders (W):** High blood pressure, high cholesterol, age, sex, and income.
- **Link:** [UCI Machine Learning Repository - CDC Diabetes Health Indicators](#) (Open Access - also widely available in Kaggle formats).

4. Pima Indians Diabetes Database

While slightly older and smaller (768 instances), this is the "Hello World" dataset for diabetes machine learning.

- **Why you should use it:** Your group of 4 should use this dataset in **August** to build your first "proof-of-concept" causal graph and test your EconML/DoWhy pipelines. It compiles quickly and is easy to debug. Once your pipeline works, swap this out for the 130-Hospitals dataset for your final December submission.
- **Link:** Available natively via Kaggle or the [National Institute of Diabetes and Digestive and Kidney Diseases \(NIDDK\)](#).



Task Distribution for the Data Team

Since you have a group of 4, here is how you divide the data engineering for the Causal Engine side of the project over the next month:

- **Member 1 (Data Ingestion & Cleaning):** Download the *130-US Hospitals* dataset. Clean the missing values (e.g., mapping "?" to null) and one-hot encode the categorical medication columns.
- **Member 2 (Causal Discovery):** Use a library like causal-learn or networkx to map out the Directed Acyclic Graph (DAG). This defines mathematically *what causes what* (e.g., Age -> BMI -> Insulin Resistance).
- **Member 3 (Treatment Modeling):** Use DoWhy to formulate the causal mechanism. Define the treatment variables (e.g., Insulin dosage) and isolate the confounders to prevent biased AI outputs.
- **Member 4 (Estimation & Counterfactuals):** Use EconML (Microsoft's causal library) to run the actual counterfactual queries. Build the API endpoint that takes in a patient profile and outputs: *"If this patient takes Metformin, the expected HbA1c drop is X%."*